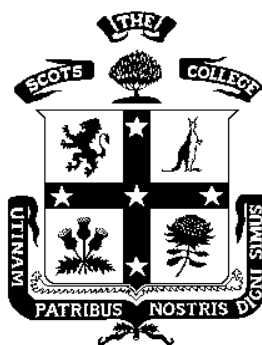

THE SCOTS COLLEGE



MATHEMATICS EXTENSION II

YEAR 12 PRETRIAL

26TH MARCH 2014

GENERAL INSTRUCTIONS

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in Section II

WEIGHTING

30%

TOTAL MARKS

70

SECTION I (5 MARKS)

- Answers to be recorded on the multiple choice answer sheet provided
- Clearly label your answer sheet with your student number
- Allow about 10 minutes for this section

SECTION II (65 MARKS)

- Questions 6 - 9
- Answers to be recorded in the answer booklets provided
- *Each question must be completed in a new answer booklet.*
- Label each answer booklet with your student number and the question number attempted. Clearly indicate the booklet order if more than one booklet is used for a question. E.g. Book 1 of 2 and 2 of 2.

SECTION I

QUESTION 1

Consider a polynomial $P(x)$ of degree 3 and the two real numbers, a and b , such that the following three conditions hold true:

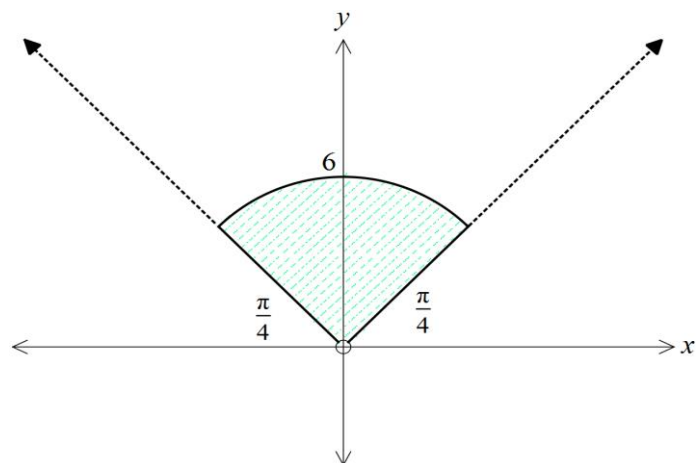
- 1) $a < b$
- 2) $P(a) > P(b) > 0$
- 3) $P'(a) = P'(b) = 0$

The polynomial $P(x)$ has:

- (A) 3 real zeros
- (B) 1 real zero γ such that $\gamma < a$
- (C) 1 real zero γ such that $a < \gamma < b$
- (D) 1 real zero γ such that $\gamma > b$

QUESTION 2

The shaded region of the Argand plane below represents the points where which of the following inequalities hold simultaneously?



- (A) $|z| \leq 6$ and $-\frac{\rho}{4} \leq \arg(z) \leq \frac{\rho}{4}$
- (B) $|z| \leq 6$ and $\frac{\rho}{4} \leq \arg(z) \leq \frac{3\rho}{4}$
- (C) $|z| \geq \sqrt{6}$ and $-\frac{\rho}{4} \leq \arg(z) \leq \frac{\rho}{4}$
- (D) $|z| \geq \sqrt{6}$ and $\frac{\rho}{4} \leq \arg(z) \leq \frac{3\rho}{4}$

QUESTION 3

The graph $f(x) = \frac{1}{x^2 + mx - n}$, where m and n are real constants, has no vertical asymptotes if:

- (A) $m^2 < -4n$
- (B) $m^2 > -4n$
- (C) $m^2 < 4n$
- (D) $m^2 > 4n$

QUESTION 4

The complex number ω is a root of the equation $z^3 + 1 = 0$. Which of the following is FALSE?

- (A) $\bar{\omega}$ is also a root.
- (B) $\omega^2 + 1 - \omega = 0$
- (C) $\frac{1}{\omega}$ is also a root.
- (D) $(\omega - 1)^2 = -1$

QUESTION 5

Given the equation of the hyperbola, $\mathcal{H}$, is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has eccentricity e , then the ellipse, $\mathcal{E}$,

with equation $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$ has which of the following as its eccentricity?

- (A) $-e$
- (B) $\frac{1}{e}$
- (C) $\sqrt{e}$
- (D) e^2

END OF SECTION I

SECTION II

QUESTION 6

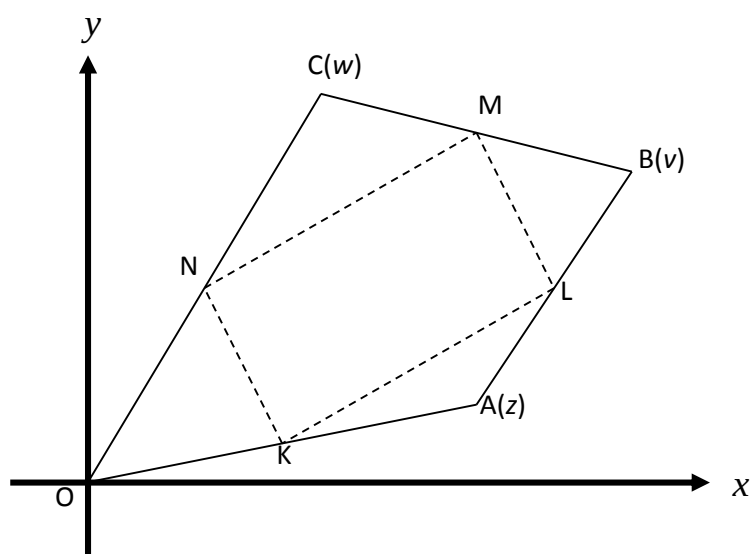
(START A NEW ANSWER BOOKLET)

21 MARKS

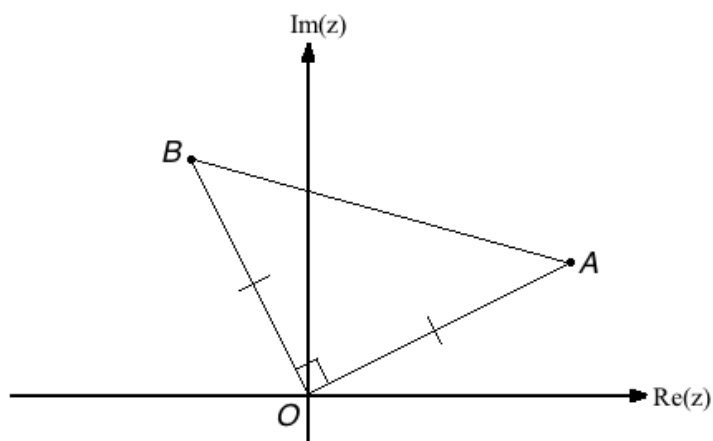
- a) Consider the complex number $z = x + iy$, where x and y are real, and the complex number $w = iz + \bar{z}$.
- i) Find w in terms of x and y and hence show that $\operatorname{Re}(w) = \operatorname{Im}(w)$. [2]
 - ii) Find all possible values of the principal argument of w . [1]
 - iii) Given that x and y are both positive and $x < y$, draw vectors representing z , iz , $\bar{z}$ and w on an Argand diagram. [2]
- b) Let $z = 12\operatorname{cis}\frac{13\pi}{15}$ and $w = 3\operatorname{cis}\frac{\pi}{5}$
- i) Evaluate $\frac{z}{w}$ and express your answer in the form $a + ib$. [2]
 - ii) Explain why $\frac{z^3}{w^3}$ is a real number. [2]
- c) The locus of z is the region simultaneously satisfied by the following conditions:
- $$|z - i| \leq 2 \quad \& \quad \operatorname{Im}(z) \leq 1$$
- i) Draw a neat sketch of the locus on an Argand diagram. [2]
 - ii) Find the values of z for which $|z|$ is a maximum and justify your answer. [2]

QUESTION 6 CONTINUES ON THE FOLLOWING PAGE...

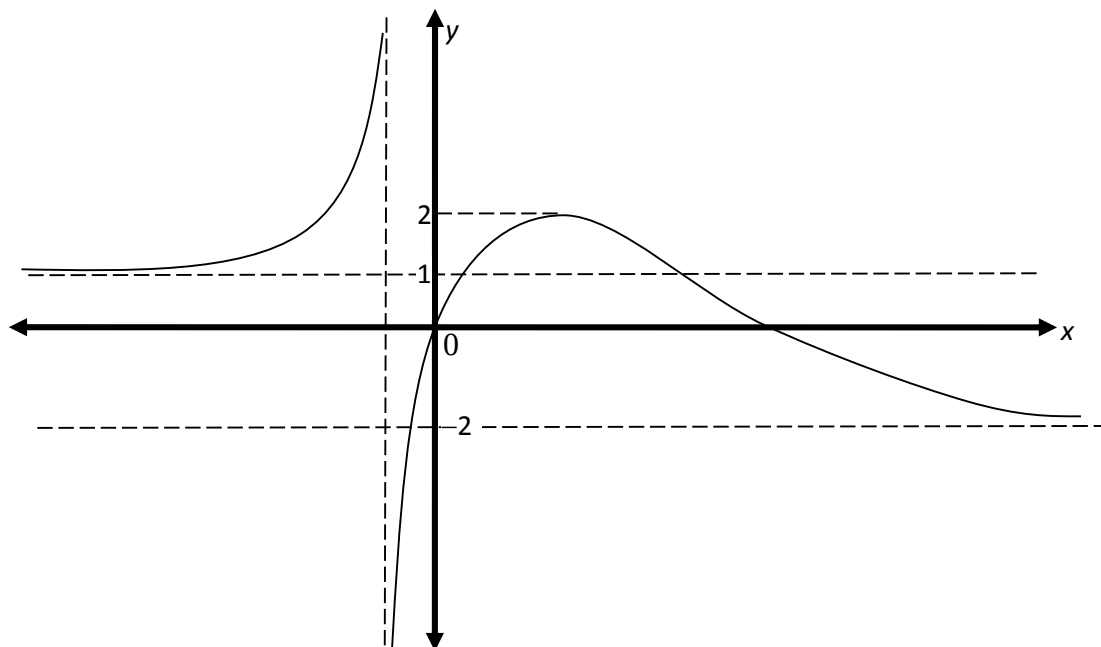
- d) OABC is a quadrilateral in the complex plane. The complex numbers, 0, z , v and w are represented by O, A, B and C respectively. K, L, M and N are the midpoints of OA, AB, BC and CO respectively.



- i) Show that the complex number represented by L is $\frac{v+z}{2}$. [2]
 - ii) Find similar results for K, M and N and hence calculate the complex numbers representing the vectors $\overrightarrow{NM}$ and $\overrightarrow{KL}$ and show that they are equal. [2]
 - iii) What type of quadrilateral is KLMN? Justify your answer. [1]
- e) The Argand diagram below shows the points A and B, which represent the complex numbers z_1 and z_2 respectively. Given that $\triangle BOA$ is a right-angled, isosceles triangle, prove that $(z_1 + z_2)^2 = 2z_1z_2$. [3]



- a) The diagram below shows the graph of the function $y = f(x)$.



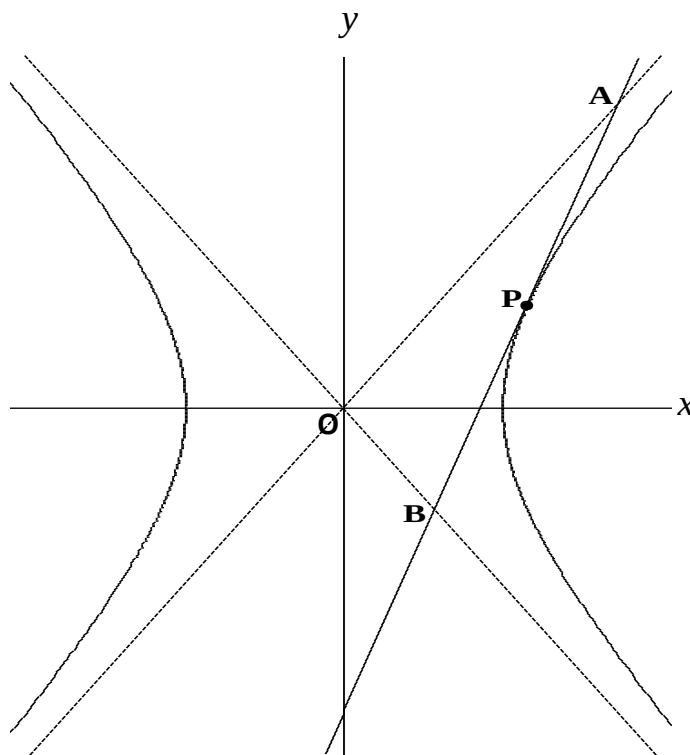
Draw a one half-page sketch for each of the following. Make sure you indicate all significant features.

- i) $y = f(-x)$ [2]
 - ii) $y = |f(x)|$ [2]
 - iii) $y = \frac{1}{f(x)}$ [2]
 - iv) $y = [f(x)]^2$ [2]
 - v) $y^2 = f(x)$ [2]
- b) Consider the curve $x^3 + 3x^2y - 2y^3 = 16$
- i) Show that $\frac{dy}{dx} = \frac{x^2 + 2xy}{2y^2 - x^2}$ [2]
 - ii) Find the coordinates of the stationary points on the curve. You do not need to determine the nature of the stationary points. [2]

- a) Consider the polynomial $P(x) = x^6 - 6x^4 + 5x^2 + 12$
- Show that i is a zero of $P(x)$ and explain why $-i$ is also a zero of $P(x)$. [2]
 - Explain why $(x^2 + 1)$ is a factor of $P(x)$. [1]
 - Given that $x^4 - 7x^2 + 12 = (x^2 - 3)(x^2 - 4)$, factorise $P(x)$ over: [3]
 - the rational field $\mathbf{Q}$
 - the real field $\mathbf{R}$
 - the complex field $\mathbf{C}$
- b) The polynomial $P(x) = x^4 - 3x^3 + 7x^2 + ax + b$ has real coefficients.
- Given that 2 is a multiple zero of $P(x)$, find the values of a and b . [2]
 - Explain why 2 is a double zero of $P(x)$ [1]

- a) The equation of an ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$.
- Show that the eccentricity of the ellipse is $\frac{4}{5}$ and determine the foci and directrices of the ellipse. [3]
 - Draw a neat sketch of the ellipse showing significant features. [2]
- b) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$. The tangents at the points P and Q meet at the point T.
- Show that the equation of the tangent at P is $x + p^2y = 2cp$. [2]
 - Show that the coordinates of T are $\left(\frac{2pqc}{p+q}, \frac{2c}{p+q}\right)$. [2]
 - If $q = 3p$, show that the locus of T is a rectangular hyperbola and find its foci. [3]

- c) The equation of the hyperbola, $\mathcal{H}$, is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The tangent at the point P with coordinates $(a \sec \theta, b \tan \theta)$ on $\mathcal{H}$ cuts the asymptotes at A and B as shown in the diagram below.



- i) Write down the equations of the asymptotes. [1]
- ii) Show that the equation of AP is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ [3]
- iii) Show that AP crosses the x-axis at the point $\left(\frac{a}{\sec \theta}, 0\right)$. [1]
- iv) Show that A is the point $\left(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta}\right)$. [2]
- v) Given that B is the point $\left(\frac{a}{\sec \theta + \tan \theta}, -\frac{b}{\sec \theta + \tan \theta}\right)$, find the exact area of $\triangle OAB$. [2]

END OF SECTION II. END OF EXAM.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



THE SCOTS COLLEGE-2014-MATHEMATICS EXTENSION 2
MATHEMATICS PRE-TRIAL HSC

CANDIDATE NUMBER: _____

SECTION I – MULTIPLE CHOICE ANSWER SHEET (5 MARKS)

Mark the correct answer by filling in the circle. To make a correction, neatly place a cross over the circle and then fill in the correct circle.

EXAMPLE:

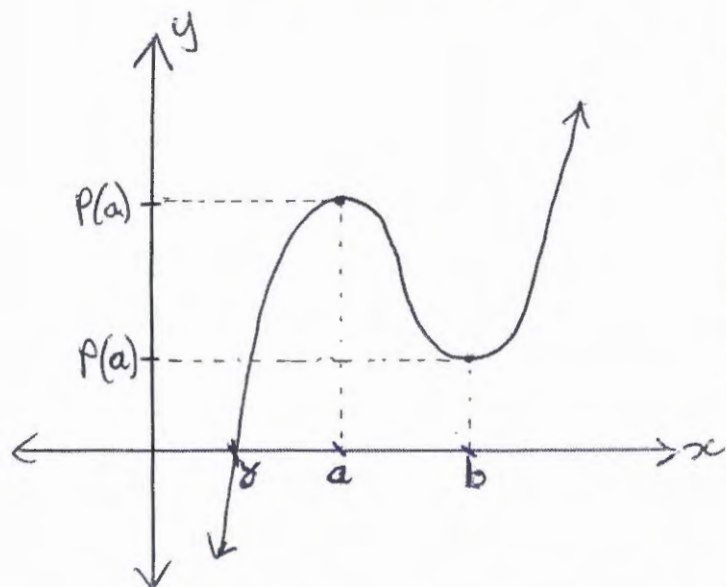
A	B	C	D
☐	☒	☒	☐

	A	B	C	D
Question 1	☐	☐	☐	☐
Question 2	☐	☐	☐	☐
Question 3	☐	☐	☐	☐
Question 4	☐	☐	☐	☐
Question 5	☐	☐	☐	☐

Mathematics Extension 2 - Pretrial 2014

Section 1:

① B: $P(x)$ has 1 real zero γ such that $\gamma < \alpha$



$a < b$, $P(a) > P(b) > 0$ and $P'(a) = P'(b) = 0$

② B: $|z| \leq 6$ and $\frac{\pi}{4} \leq \arg z \leq \frac{3\pi}{4}$

③ A: $f(x) = \frac{1}{x^2 + mx - n}$

For $f(x)$ to have no vertical asymptotes, we need $\Delta < 0$ for $x^2 + mx - n$

$$\Delta < 0$$

$$m^2 - 4(1)(-n) < 0$$

$$m^2 + 4n < 0$$

$$m^2 < -4n$$

④ D: $z^3 + 1 = 0$
 $z^3 = -1$

$$z^3 = \cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi)$$

for $k = 0, 1, 2$

$$z = \cos\left(\frac{\pi + 2k\pi}{3}\right) + i\sin\left(\frac{\pi + 2k\pi}{3}\right)$$

$$= \cos \frac{\pi}{3} + i\sin \frac{\pi}{3},$$

$$\cos \pi + i\sin \pi \text{ or}$$

$$\cos \frac{5\pi}{3} + i\sin \frac{5\pi}{3}$$

$$= \cos \frac{\pi}{3} + i\sin \frac{\pi}{3}, -1 \text{ or } \cos \frac{\pi}{3} - i\sin \frac{\pi}{3}$$

$\therefore \bar{w}$ is a root

$\frac{1}{w}$ is a root

$$\text{Also, } w^3 + 1 = 0$$

$$(w+1)(w^2 - w + 1) = 0$$

$$\text{So } w^2 + 1 - w = 0$$

$$\begin{aligned} \text{But if } w = -1, (w-1)^2 &= (-1-1)^2 \\ &= (-2)^2 \\ &= 4 \\ &\neq -1 \end{aligned}$$

So D is false

⑤ B: For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

we have $b^2 = a^2(e^2 - 1)$

$$\frac{b^2}{a^2} = e^2 - 1$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$= \frac{a^2 + b^2}{a^2}$$

$$e = \frac{\sqrt{a^2 + b^2}}{a}$$

For the ellipse $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$

we have $b^2 = (a^2 + b^2)(1 - E^2)$

$$\frac{b^2}{a^2 + b^2} = 1 - E^2$$

$$E^2 = 1 - \frac{b^2}{a^2 + b^2}$$

$$= \frac{a^2 + b^2 - b^2}{a^2 + b^2}$$

$$= \frac{a^2}{a^2 + b^2}$$

$$E = \frac{a}{\sqrt{a^2 + b^2}}$$

$$= \frac{1}{e}$$



Alternatively:

Hyperbolas have eccentricity e such that $e > 1$

$$\text{So } -e < 0$$

$$\frac{1}{e} < 1$$

$$\sqrt{e} > 1$$

$$\text{and } e^2 > 1$$

Ellipses have eccentricity e such that $0 < e < 1$

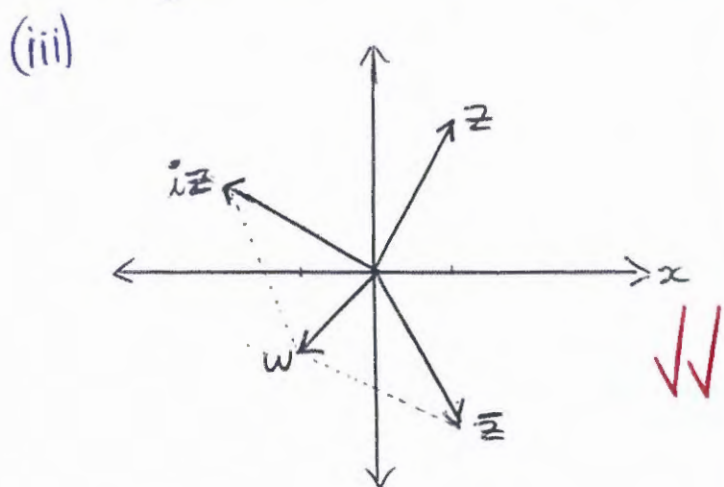
$\therefore \frac{1}{e}$ is the ellipse eccentricity

5

⑥ Complex Numbers

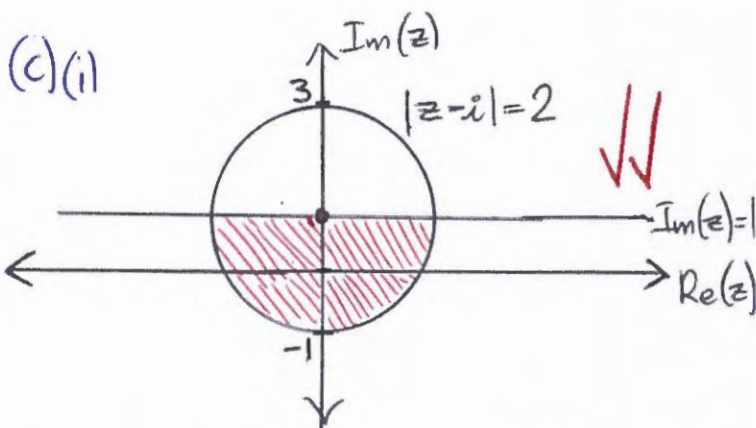
(a)(i) $w = iz + \bar{z}$
 $= i(x+iy) + x-iy$
 $= ix - y + x - iy$
 $= x - y + i(x - y)$ ✓
 $\text{Re}(w) = x - y$ ✓
 $= \text{Im}(w)$ ✓

(ii) If $\text{Re}(w) = \text{Im}(w)$, then
 $\arg w = \frac{\pi}{4} \text{ or } -\frac{3\pi}{4}$ ✓



(b)(i) $\frac{z}{w} = \frac{12 \text{ cis } \frac{13\pi}{15}}{3 \text{ cis } \frac{\pi}{5}}$
 $= \frac{12}{3} \times \text{cis} \left(\frac{13\pi}{15} - \frac{\pi}{5} \right)$
 $= 4 \times \text{cis} \left(\frac{13\pi}{15} - \frac{3\pi}{15} \right)$
 $= 4 \text{ cis} \left(\frac{10\pi}{15} \right)$ ✓
 $= 4 \text{ cis } \frac{2\pi}{3}$
 $= 4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$
 $= 4 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$ ✓
 $= -2 + 2\sqrt{3}i$ ✓

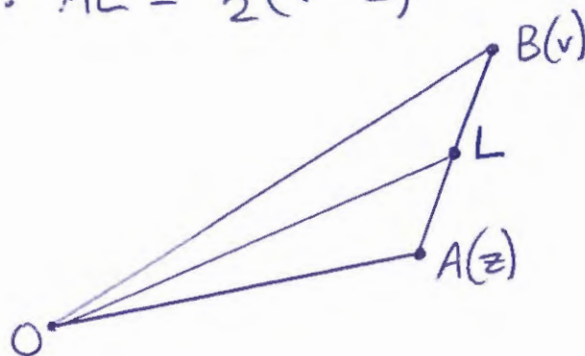
(ii) $\frac{z^3}{w^3} = \left(\frac{z}{w} \right)^3$
 $= \left(4 \text{ cis } \frac{2\pi}{3} \right)^3$
 $= 4^3 \times \text{cis} \left(\frac{3 \times 2\pi}{3} \right)$ ✓
 $= 64 \text{ cis } 2\pi$
 $= 64 (\cos 2\pi + i \sin 2\pi)$
 $= 64 (1 + 0i)$ ✓
 $= 64$ which is a real number ✓



(ii) The points in the region furthest from the origin are where the line $\text{Im}(z) = -1$ intersects the circle, so $|z|$ would be at a maximum value

$\therefore z = -2 + i \text{ or } 2 + i$ ✓✓

(d)(i) $AL = \frac{1}{2} AB$
 because L is the midpoint of AB
 $\therefore AL = \frac{1}{2}(v - z)$ ✓



$$\begin{aligned}
 OL &= OA + AL \text{ (vector addition)} \\
 &= z + \frac{1}{2}(v-z) \\
 &= \frac{2z + v - z}{2} \\
 &= \frac{v+z}{2}
 \end{aligned}$$

So L is represented by the complex number $\frac{v+z}{2}$

(ii) Similarly, K, M and N are represented by the complex numbers

$$\frac{z}{2}, \frac{v+w}{2} \text{ and } \frac{w}{2} \text{ respectively}$$

$$\therefore \vec{NM} = \frac{v+w}{2} - \frac{w}{2}$$

$$= \frac{v}{2}$$

$$\vec{KL} = \frac{v+z}{2} - \frac{z}{2}$$

$$= \frac{v}{2}$$

$$\text{So } \vec{NM} = \vec{KL}$$

(iii) Because $\vec{NM} = \vec{KL}$, then

$NM = KL$ (in length) and

$NM \parallel KL$ (parallel)

So KLMN is a parallelogram because it has an equal pair of opposite sides which are parallel

(e) Since $\triangle BOA$ is right-angled and isosceles, we have

$$z_2 = iz_1$$

$$\therefore (z_1 + z_2)^2 = (z_1 + iz_1)^2$$

$$= z_1^2 + 2iz_1^2 + i^2 z_1^2$$

$$= z_1^2 + 2iz_1^2 - z_1^2$$

$$= 2iz_1^2$$

$$= 2 \times z_1 \times iz_1$$

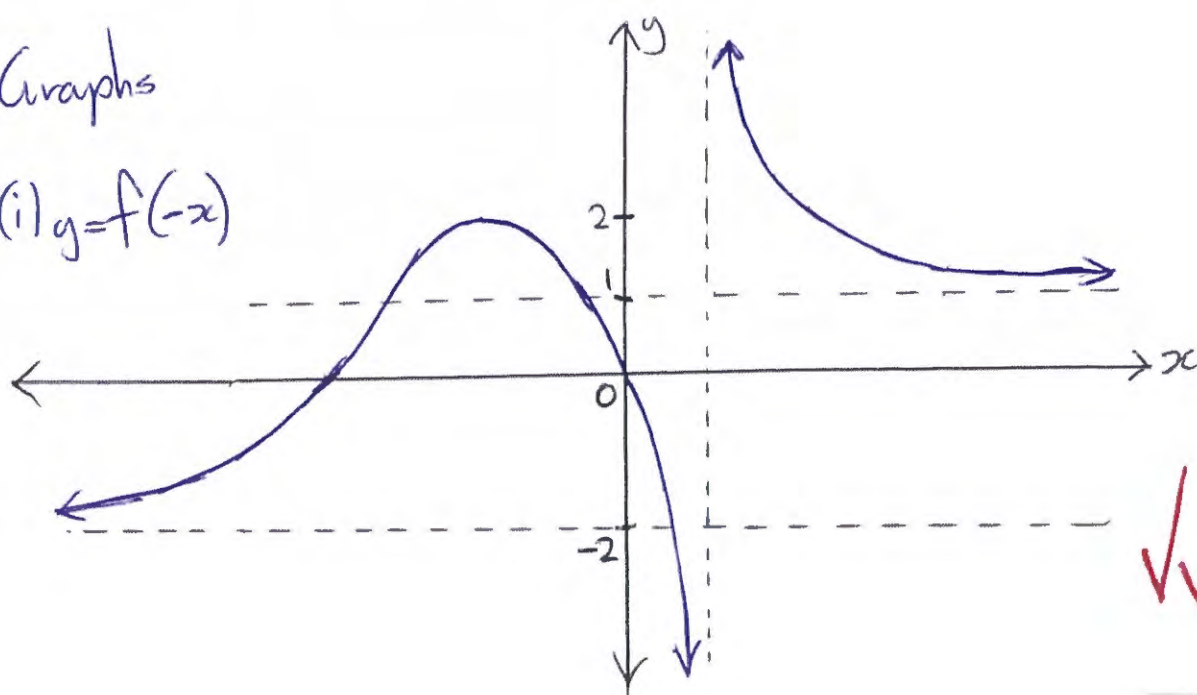
$$= 2 \times z_1 \times z_2$$

$$= 2z_1 z_2$$

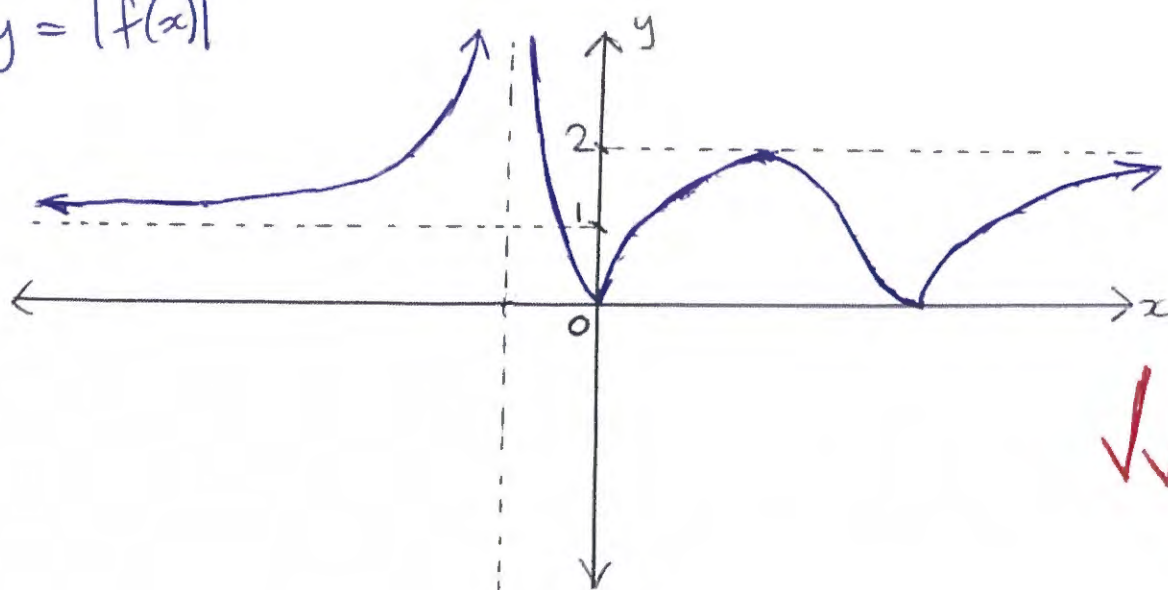
21

7 Graphs

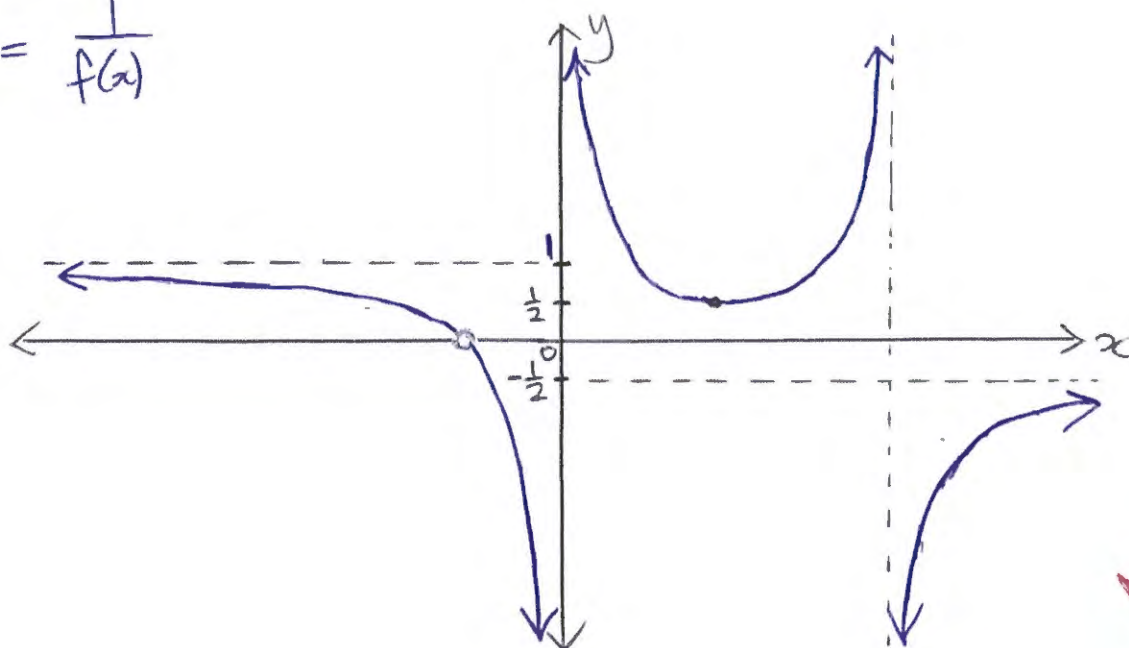
(a) (i) $y = f(-x)$



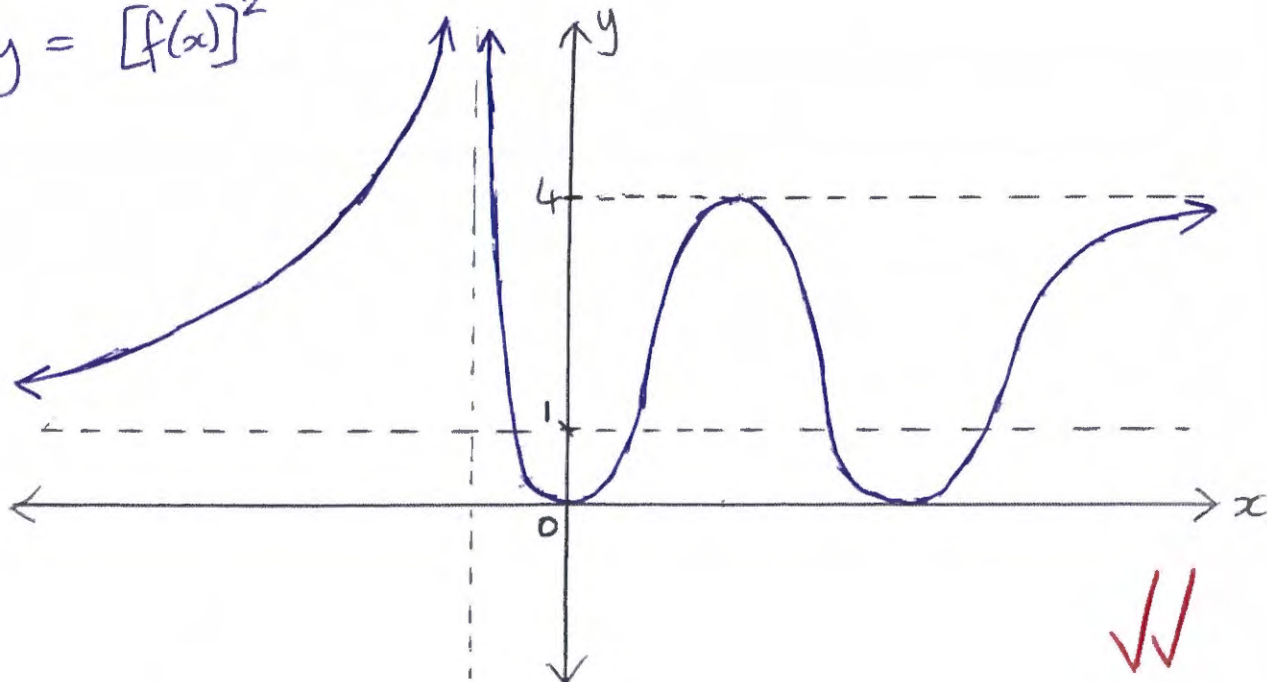
(ii) $y = |f(x)|$



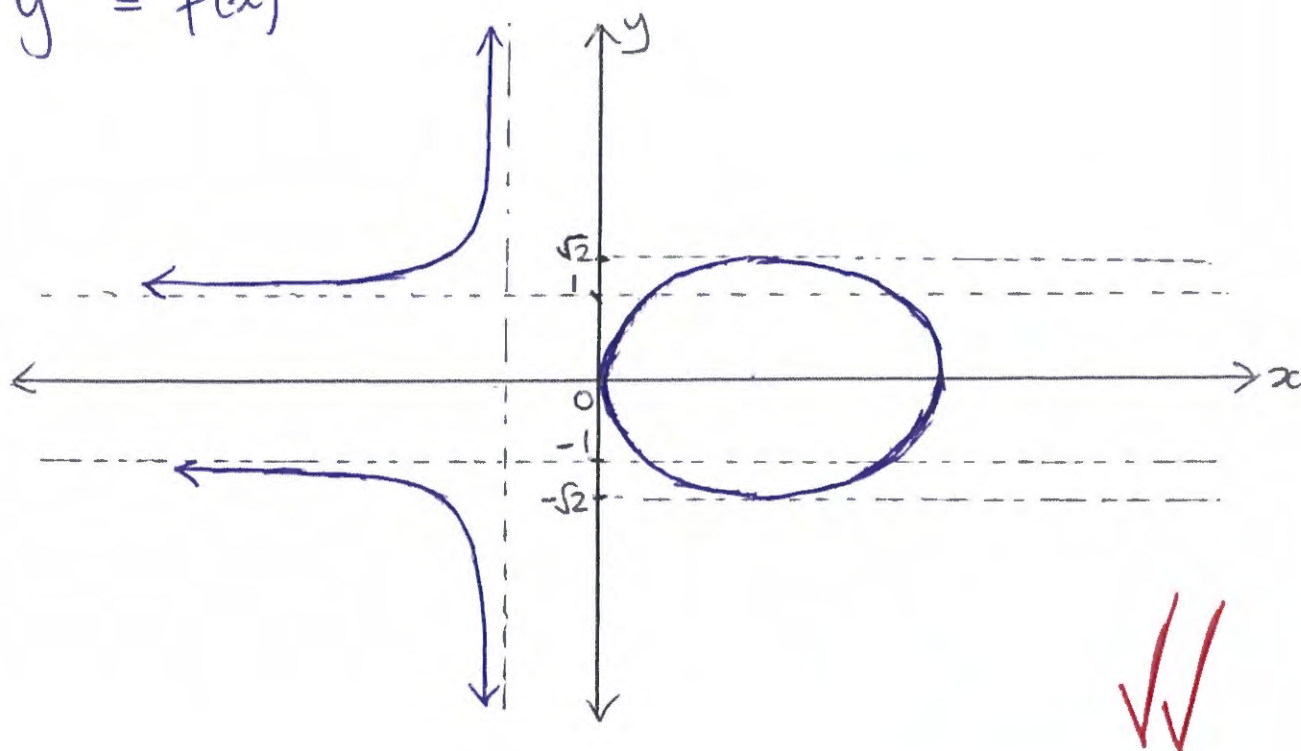
(iii) $y = \frac{1}{f(x)}$



(iv) $y = [f(x)]^2$



(v) $y^2 = f(x)$



(b)(i) $x^3 + 3x^2y - 2y^3 = 16$

$$\frac{d}{dx}(x^3 + 3x^2y - 2y^3) = \frac{d}{dx}(16)$$

$$3x^2 + 6xy + 3x^2 \frac{dy}{dx} - 6y^2 \frac{dy}{dx} = 0$$

$$x^2 + 2xy + x^2 \frac{dy}{dx} - 2y^2 \frac{dy}{dx} = 0$$

✓

$$\frac{dy}{dx}(x^2 - 2y^2) = -x^2 - 2xy$$

$$\frac{dy}{dx} = \frac{-(x^2 + 2xy)}{x^2 - 2y^2}$$

$$= \frac{x^2 + 2xy}{2y^2 - x^2}$$

(ii) $\frac{dy}{dx} = 0$ for stationary points

$$x^2 + 2xy = 0$$

$$x(x + 2y) = 0$$

$$x = 0 \text{ or } -2y$$

$$\text{When } x = 0, \quad (0)^3 + 3(0)^2y - 2y^3 = 16$$

$$-2y^3 = 16$$

$$y^3 = -8$$

$$y = -2$$

$\therefore (0, -2)$ is a stationary point

$$\text{Also, when } x = -2y, \quad (-2y)^3 + 3(-2y)^2y - 2y^3 = 16$$

$$-8y^3 + 12y^3 - 2y^3 = 16$$

$$2y^3 = 16$$

$$y^3 = 8$$

$$y = 2$$

$$\text{When } y = 2, \quad x = -2(2)$$

$$= -4$$

$\therefore (-4, 2)$ is a stationary point

So there are stationary points at $(0, -2)$ and $(-4, 2)$

POLYNOMIALS

$$\textcircled{8} (a) \quad P(i) = (i)^6 - 6(i)^4 + 5(i)^2 + 12$$

$$= -1 - 6 - 5 + 12$$

$$= 0$$

$\therefore i$ is a zero of $P(x)$

Since the coefficients of $P(x)$ are all real numbers, the conjugate of i , which is $-i$, is also a zero because the complex roots of $P(x)$ must occur in conjugate pairs

Check: $P(-i) = (-i)^6 - 6(-i)^4 + 5(-i)^2 + 12$

$$= -1 - 6 - 5 + 12$$

$$= 0$$

(ii) Both $(x-i)$ and $(x+i)$ are factors of $P(x)$ so we have

$$(x-i)(x+i) = x^2 + 1$$

which is a factor of $P(x)$

(iii) (a) $P(x) = (x^2+1)(x^2-3)(x^2-4)$

$$= (x^2+1)(x^2-3)(x+2)(x-2)$$

over the rational field $\mathbb{Q}$

(B) $P(x) = (x^2+1)(x+\sqrt{3})(x-\sqrt{3})(x+2)(x-2)$

over the real field $\mathbb{R}$

(C) $P(x) = (x+i)(x-i)(x+\sqrt{3})(x-\sqrt{3})(x+2)(x-2)$

over the complex field $\mathbb{C}$

(b)(i) $P(x) = x^4 - 3x^3 + 7x^2 + ax + b$

$$P'(x) = 4x^3 - 9x^2 + 14x + a$$

$$P'(2) = 4(2)^3 - 9(2)^2 + 14(2) + a$$

$$0 = 32 - 36 + 28 + a$$

$$a = -24$$

Also, $P(2) = 0$

$$0 = 2^4 - 3(2)^3 + 7(2)^2 - 24(2) + b$$

$$0 = 16 - 24 + 28 - 48 + b$$

$$b = 28$$

$$\therefore P(x) = x^4 - 3x^3 + 7x^2 - 24x + 28$$

(ii) $P'(x) = 4x^3 - 9x^2 + 14x$

$$P''(x) = 12x^2 - 18x + 14$$

However, $P''(2) = 12(2)^2 - 18(2) + 14$

$$= 48 - 36 + 14$$

$$= 26$$

$$\neq 0$$

Because $P(2) = 0$ and $P'(2) = 0$ but $P''(2) \neq 0$, 2 is a double root of $P(x)$

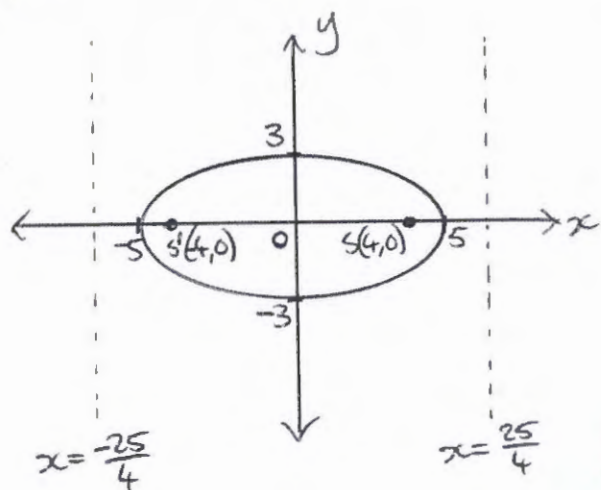
9

(a) (i) $b^2 = a^2(1-e^2)$
 $9 = 25(1-e^2)$
 $\frac{9}{25} = 1-e^2$
 $e^2 = \frac{16}{25}$
 $e = \sqrt{\frac{16}{25}}$
 $= \frac{4}{5}$

foci are $(\pm ae, 0)$
 $= (\pm 5 \times \frac{4}{5}, 0)$
 $= (-4, 0) \text{ and } (4, 0)$

directrices are $x = \pm \frac{a}{e}$
 $= \pm \frac{5}{\frac{4}{5}}$
 $= \pm 5 \times \frac{5}{4}$
 $= -\frac{25}{4} \text{ and } \frac{25}{4}$

(ii)



$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

(b) (i) $xy = c^2$
 $y = \frac{c^2}{x}$
 $= c^2 x^{-1}$
 $\frac{dy}{dx} = -\frac{c^2}{x^2}$
At $P(cp, \frac{c}{p})$, $\frac{dy}{dx} = -\frac{c^2}{(cp)^2}$
 $= -\frac{c^2}{c^2 p^2}$
 $= -\frac{1}{p^2}$

$\therefore y - y_1 = m(x - x_1)$
 $y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp$$

is the equation of the tangent at P

(ii) The tangent at Q has equation

$$x + q^2 y = 2cq$$

So we have

$$x + p^2 y = 2cp \quad (1)$$

$$x + q^2 y = 2cq \quad (2)$$

(1) - (2):

$$(p^2 - q^2)y = 2c(p - q)$$

$$(p+q)(p-q)y = 2c(p-q)$$

$$y = \frac{2c}{p+q}$$

When $y = \frac{2c}{p+q}$

$$\begin{aligned} x + p^2 \left(\frac{2c}{p+q} \right) &= 2cp \\ x &= 2cp - \frac{2cp^2}{p+q} \\ &= \frac{2cp(p+q) - 2cp^2}{p+q} \\ &= \frac{2cp^2 + 2pqc - 2cp^2}{p+q} \\ &= \frac{2pqc}{p+q} \quad \checkmark \end{aligned}$$

So T is the point $\left(\frac{2pqc}{p+q}, \frac{2c}{p+q} \right)$

(iii) If $q = 3p$

$$\begin{aligned} T &= \left(\frac{2p(3p)c}{p+3p}, \frac{2c}{p+3p} \right) \\ &= \left(\frac{6p^2c}{4p}, \frac{2c}{4p} \right) \\ &= \left(\frac{3pc}{2}, \frac{c}{2p} \right) \quad \checkmark \end{aligned}$$

If $y = \frac{c}{2p}$
 $p = \frac{c}{2y}$

and $x = \frac{3pc}{2}$

$$\begin{aligned} &= \frac{3c}{2} \times \frac{c}{2y} \\ &= \frac{3c^2}{4y} \end{aligned}$$

$\therefore xy = \frac{3c^2}{4} \quad \checkmark$

This is a rectangular hyperbola

$$\begin{aligned} xy &= \left(\frac{\sqrt{3}c}{2} \right)^2 \\ &= d^2 \text{ for } d = \frac{\sqrt{3}c}{2} \end{aligned}$$

Foci of a rectangular hyperbola are at the points $(\pm a, \pm a)$

$$\begin{aligned} &= (\pm \sqrt{2}d, \pm \sqrt{2}d) \\ &= \left(\pm \sqrt{2} \times \frac{\sqrt{3}c}{2}, \pm \sqrt{2} \times \frac{\sqrt{3}c}{2} \right) \\ &= \left(-\frac{\sqrt{6}c}{2}, -\frac{\sqrt{6}c}{2} \right) \text{ and } \left(\frac{\sqrt{6}c}{2}, \frac{\sqrt{6}c}{2} \right) \quad \checkmark \end{aligned}$$

(c) (i) Asymptotes: $y = -\frac{bx}{a}$ and $y = \frac{bx}{a} \quad \checkmark$

(ii) $\frac{d}{dx} \left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right) = \frac{d}{dx} (1)$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

At $P(a \sec \theta, b \tan \theta)$;

$$\frac{dy}{dx} = \frac{b^2}{a^2} \times \frac{a \sec \theta}{b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta} \quad \checkmark$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$ab \sec^2 \theta - ab \tan^2 \theta = bx \sec \theta - ay \tan \theta$$

$$ab(\sec^2 \theta - \tan^2 \theta) = bx \sec \theta - ay \tan \theta$$

$$ab = bx \sec \theta - ay \tan \theta$$

$$1 = \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b}$$

(iii) When $y = 0$,

$$1 = \frac{x \sec \theta}{a}$$

$$a = x \sec \theta$$

$$x = \frac{a}{\sec \theta}$$

So AP crosses the x-axis at the point $(\frac{a}{\sec \theta}, 0)$

(iv) At A, we have $y = \frac{bx}{a}$

$$\therefore \frac{x \sec \theta}{a} - \left(\frac{bx}{a}\right) \frac{\tan \theta}{b} = 1$$

$$\frac{x \sec \theta}{a} - \frac{x \tan \theta}{a} = 1$$

$$x(\sec \theta - \tan \theta) = a$$

$$x = \frac{a}{\sec \theta - \tan \theta}$$

$$y = \frac{b}{a} x$$

$$= \frac{b}{a} \times \frac{a}{\sec \theta - \tan \theta}$$

$$= \frac{b}{\sec \theta - \tan \theta}$$

$\therefore$ A is the point $(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta})$

(v) Area ΔOAB = Area ΔOAX + Area ΔOBX

Taking OX as the base length of each triangle, we have

$$\text{Area} = \frac{1}{2} \times \text{base} \times (\text{height}_1 + \text{height}_2)$$

$$= \frac{1}{2} \times \frac{a}{\sec \theta} \times \left(\frac{b}{\sec \theta - \tan \theta} + \frac{b}{\sec \theta + \tan \theta} \right)$$

$$= \frac{a}{2 \sec \theta} \left(\frac{b(\sec \theta + \tan \theta) + b(\sec \theta - \tan \theta)}{\sec^2 \theta - \tan^2 \theta} \right)$$

$$= \frac{a}{2 \sec \theta} \times \frac{2b \sec \theta}{1}$$

$$= \frac{2ab \sec \theta}{2 \sec \theta}$$

$$= ab \text{ units}^2$$

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